

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Fifth Semester B.E. (CSE) (Credit System) Degree Examinations
 January 2023

20CSE71 - ARTIFICIAL INTELLIGENCE

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Define Artificial Intelligence (AI) and briefly list out the applications of AI. | 10 | L*2 | 1 | 1 |
| b) Explain and apply simple reflex agent program for vacuum cleaner and also explain the problem that would arise in the vacuum world if the vacuum agent is deprived of its location sensor and dirt sensor. | 10 | L3 | 1 | 1 |
| 2. a) Develop the PEAS description for Mushroom picking robot and automated taxi. | 10 | L3 | 1 | 2 |
| b) Summarize general tree-search and graph - search algorithms. | 10 | L2 | 2 | 1 |
| 3. a) Construct and explain uninformed search strategy-Breadth First Search algorithm on a graph. | 10 | L3 | 2 | 1 |
| b) Formulate for 8-Queens problem with all the necessary components for problem formulation. | 10 | L3 | 2 | 2 |

Unit – II

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|---|----|----|---|---|
| 4. a) Summarize the reasons for reasoning under uncertainty and explain probabilistic reasoning with example. | 10 | L2 | 3 | 1 |
| b) Discuss Bayes rule. Apply and explain its use for casual diagnostic. | 10 | L3 | 3 | 2 |
| 5. a) Construct the pseudo code structure of an agent that uses decision theory to select actions. Explain the principle of Maximum Expected Utility. | 10 | L3 | 3 | 1 |
| b) Given below the full joint distribution of the dental problem involving Toothache, Cavity, Catch world. | | | | |

	toothache		¬toothache	
	catch	¬ catch	catch	¬ catch
cavity	0.108	0.012	0.072	0.008
¬cavity	0.106	0.064	0.144	0.576

Solve the following: i) $P(\text{cavity} \vee \text{toothache})$ ii) marginal probability of cavity iii) probability of a cavity, given evidence of a toothache iv) the probability that there is no cavity, given evidence of a toothache

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|--|----|----|---|---|
| | 10 | L3 | 4 | 2 |
| 6. a) Discuss about forward backward algorithm for smoothing. | 10 | L2 | 4 | 1 |
| b) Formulate 5 basic inference tasks in temporal models. | 10 | L3 | 4 | 2 |
| Unit – III | | | | |
| 7. a) Explain reinforcement learning with example. Make use of it and explain three approaches for learning utilities. | 10 | L3 | 5 | 2 |
| b) Outline the algorithm for exploratory Q-Learning agent. | 10 | L2 | 5 | 1 |
| 8. a) Explain the application of reinforcement learning in Gaming. | 10 | L3 | 5 | 2 |
| b) Write short notes on i) Generalization ii) Policy Search. | 10 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
